Course architecture

Advanced Statistics 2027-1 / Probabilistic Machine Learning · Course-level intellectual and curricular authority

This document defines the **intellectual purpose and planned progression** of the course. It is the primary reference when selecting lecture scopes, weighing changes across lectures, and interpreting the relationship between mathematical results and applications. The student syllabus is a concise description of this architecture; approved lecture scopes determine their own detailed teaching commitments. Actual delivery and source coverage are recorded in the material ledger. These authorities should be reconciled when they disagree, not silently overwritten.

What the course is about

Understanding probabilistic machine learning is understanding **why different classes of learning problems call for different mathematical machinery**. The point is neither a catalogue of models nor a progression from one approximation algorithm to the next. Each mathematical advance is motivated by a kind of task and available experience for which our existing models, performance criteria, or computations are insufficient.

From learning problems to mathematical formulations

L01 established the common starting point. A **learning task** specifies what a system should accomplish; its **experience** specifies the data, feedback, or interaction available. Its success must be defined by a **performance criterion**. The same dataset may support prediction, generation, discovery, or action; data alone do not specify the learning problem.

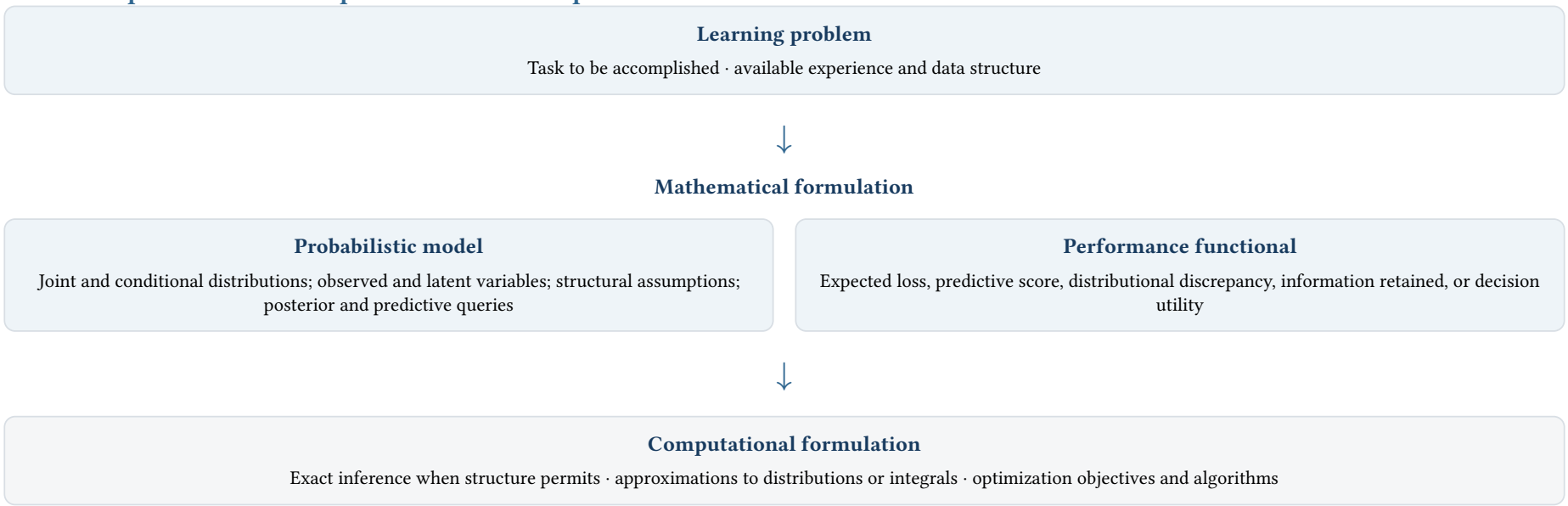
The mathematical formulation has **two distinct, related objects**. A probabilistic model describes the relevant observed and unobserved variables, their dependencies, and our uncertainty. A performance functional defines what counts as a good prediction, representation, decision, or learned distribution—often by taking an expectation with respect to a relevant distribution. For example, a predictive action a may be evaluated by

$$\mathcal{R}(a) = \int L(a, y) p(y) \, \mathrm{d}y.$$

The distribution under which we evaluate actual performance need not be the same as our fitted model. Distinguishing a mismatch of models from an error of computation is itself part of the discipline.

The formulation then gives us probabilistic **queries**. Conditioning, marginalization, and summarizing a distribution—three operations developed explicitly in L01—may deliver the quantities required to evaluate or optimize performance. They are tractable in some models, including the linear-Gaussian systems of L01, but not in every family of tasks.

The task–model–performance–computation relationship



Learned model, answer, or decision

Evaluate performance on the original task; identify remaining model, objective, and computational limitations

The arrows describe the logical dependence, not a one-way workflow. Results and computational limits can motivate revising the formulation.

The three major **moves** in the course are consequently extensions of **(i) what models can represent, (ii) what performance functionals can express, and (iii) what computation can achieve**. A task family motivates one move or a combination; no move is an end in itself.

The ideal quantity and the practical calculation

It is useful to distinguish a mathematical target from the tractable proxy used to pursue it. A desired performance functional may be replaced during training by an empirical score or surrogate objective. Likewise, an ideal posterior or expectation may be represented by a point estimate, a local Gaussian approximation, a variational distribution, or samples. Some structured problems admit **exact** solutions by message passing, so computation is not synonymous with approximation.

These substitutions have different meanings and limitations. A negative log-likelihood objective may support good distributional prediction without exactly matching task-specific performance. A posterior mode does not represent posterior spread; an exact EM step improves a marginal-likelihood objective but does not provide a full posterior over global parameters; a finite MCMC run approximates particular expectations. The appropriate criterion for assessing the approximation depends on the **original question**, not merely whether an internal algorithm converged.

The role of Murphy and mathematical depth

Murphy's books provide the principal mathematical argument, but no lecture attempts complete chapter coverage. We select and sometimes reorder developments to build a recognizable source-based argument that serves the current family of problems and connects to the preceding course. Where a practical notebook example motivates a lecture, it must not quietly replace the intellectual role of the mathematical material. A lecture should reach a substantive conclusion and make explicit what it enables and what it does not solve.

How the semester develops

Foundations: questions, distributions, and structure

L01 establishes task, experience, performance, and the recurrent probabilistic operations. Linear-Gaussian systems show how conditioning, marginalization, and summarizing can be done coherently and exactly. **L02** connects several observation models through exponential families and derives fitting by moment matching; learning categorical transitions leads to a simple sequence generator. **L03** examines factorizations and conditional independence to express larger structured joint models without writing a full joint table.

Richer performance criteria and tractable learning

L04 addresses tasks for which single-answer classification accuracy or mean squared error is inadequate. Distributional similarity, relative information, and preservation of relevant statistical relationships motivate entropy, cross-entropy, KL divergence, mutual information, and their relationship to computable scores or bounds. The selected small noisy-channel example is a **worked illustration**, not the intellectual definition of the chapter or a representative machine-learning application: it distinguishes information, decision reliability, and resource cost. The three-bit detector is live; the five-bit correction problem extends it in the notebook.

L05 connects familiar Bayesian regression to likelihood families for categorical and count observations. Exponential-family GLMs extend the **model**. Differentiation and stochastic optimization address the practical problem of **learning** from finite observations. Using MLE or MAP as a point fit is a computational choice that retains the probabilistic model but not a full parameter posterior. Prediction is ultimately judged by an appropriate performance criterion; success on an internal training objective is not equivalent to task performance.

L06 retains a familiar latent-variable hierarchy and asks how parameters can be learned when the observed-data likelihood involves marginalization over hidden values. EM uses conditional inference to construct tractable expected-statistic fitting steps, under explicit assumptions. The same probabilistic framework motivates the method; it is not a competing definition of hierarchical modeling.

Inference and computation under different structural conditions

L07 returns to the difference between a point fit and posterior queries that require uncertainty or predictive integration. Laplace uses local curvature for a Gaussian approximation; the approximation must be checked for the query that matters.

L08 addresses sequential tasks where the naive latent sum appears exponential but the joint factorization enables efficient **exact** filtering and smoothing by message passing. **L09** instead optimizes a restricted distribution to approximate an otherwise difficult posterior; **L10** uses samples to approximate integrals and studies the validity and finite-run error of Monte Carlo estimates. These methods solve different computational bottlenecks, not interchangeable names for "approximate inference."

Expressive models and learned inference

L11 develops compositional differentiable functions as components of conditional probability models. Greater expressive capacity can improve nonlinear prediction but does not itself solve parameter uncertainty. Neural networks can also parameterize **inference** maps, a role developed in **L12**. The VAE combines a generative latent-variable model, amortized posterior approximation, and an objective that trades reconstruction quality against an information-related rate. Generation and discovery are different tasks even when they share the same model.

Lecture allocation and notebook commitments

The table below retains the selected 13-meeting allocation and source readings. It is a **planning contract**: L01 is delivered, L02 is agreed, L03 preserves its instructor-approved scope, and L04′s live/notebook split is selected. Other lecture scopes remain proposed pending their own reviews. Each notebook should extend rather than reproduce the live mathematics, normally within a separate hour. Full mathematical source-to-lecture ownership and unresolved feasibility questions remain in the architecture audit and the scope register.

No.	Lecture	Topics covered	Book 2 — primary	Book 1 — supplemental	Notebook problem, worked reference and skill
1	**Probabilistic ML and Gaussian systems** Delivered	Machine-learning task framing; probabilistic models; Bayes review; covariance and correlation; multivariate Gaussians; conditioning; missing-data inference; linear Gaussian systems.	• 2.1.5–2.1.6 Conditional probability; Bayes • 2.2.1.1 Bernoulli and binomial • 2.2.4.1 Beta distribution • 2.3.1.1 Gaussian joint distribution • 2.3.1.3–2.3.1.5 Gaussian conditionals; information form • 2.3.2.1–2.3.2.2 Linear-Gaussian systems • 3.2.1 Coin-flipping Bayes • 3.4–3.4.1 Conjugate priors; binomial	• 1.1–1.5 ML tasks and datasets • 2.3 Bayes' rule • 2.4.1 Bernoulli; binomial • 2.7.4 Beta distribution • 3.1.1–3.1.3 Covariance; correlation • 3.2.1–3.3.4 Multivariate and linear Gaussians • 4.3–4.3.1 Empirical risk • 4.6.1–4.6.2 Beta-binomial Bayes	**Gaussian measurement imputation (already delivered).** Condition a multivariate Gaussian on observed measurements, infer missing values and their uncertainty. Reference: Existing L1 solved notebook.
2	**Exponential families and learning from sufficient statistics** Agreed scope	Exponential-family representation; Gaussian canonical form; derivatives of the log-partition function; natural and mean parameters; MLE as moment matching; maximum entropy; Gamma fit; categorical counts; Markov-chain learning.	• 2.2.1.2 Categorical distribution • 2.4.1–2.4.2 Exponential-family form; examples • 2.4.3–2.4.4 Log partition; moments • 2.4.5 MLE; moment matching • 2.4.7 Maximum entropy • 2.6.1–2.6.3 Markov chains; language models		**Baby-name character bigram model.** Count character transitions in the names corpus, normalize the transition matrix, compute sequence NLL and generate new names. Reference: Karpathy's solved bigram notebook.
3	**Graphical models: factorization and conditional independence** Agreed scope restored	Derive DAG factorization from the probability chain rule and independence assumptions; work the student network and parameter economy. Develop d-separation and directed Markov blankets. Define undirected potentials and global normalization, compare separation and independence expressivity, and interpret the binary Ising model. Image posterior and algorithm derivations belong to the notebook.	• 4.2.1–4.2.2 DAG factorization; student network • 4.2.4 / selected 4.2.8 Independence; plates • 4.3.1–4.3.2.1 / 4.3.6.1 Undirected models; Ising; separation • 4.5.1 Directed/undirected independence comparison • 4.3.8 Image restoration: notebook	• Optional 3.6.1–3.6.2 Graphical-model orientation	Construct an Ising prior and Gaussian observation likelihood; derive the joint, posterior and single-site conditional. Implement or complete sequential local maximization (ICM) in a NumPy scaffold; compare with a pixelwise baseline and examine initialization or local-optimum limitations. **Skill:** Build a probabilistic restoration model from local factors, derive its posterior update, and distinguish local maximization from a global optimum or posterior marginal inference. Inspected source. **Limit:** The instructor agreed to this simpler method. It differs from Murphy's algorithm and does not promise the best possible image or posterior uncertainty. All course adaptations remain unrun.
4	**Information theory: mutual information, processing and noisy coding** Proposed teaching scope	Finite entropy, cross-entropy and KL divergence; derive mutual information and sketch the data-processing inequality with its conditional-independence premise. State source-length bounds. Work three-bit parity detection and posterior decoding, distinguishing information, Bayes message risk and a coding guarantee.	• 5.1.3 / 5.1.6; 5.2.1 / 5.2.4 Information and uncertainty • 5.3.1–5.3.4 Shared information and processing • 5.4.1 Code-length bounds: statement only • 5.5 Noisy-channel example: three-bit parity model	• Selected 6.1 / 6.2.1–6.2.2 / 6.3 Uncertainty and shared information	Extend the lecture's three-bit parity detector to a five-bit correcting code for the same two-bit message. Check minimum Hamming distance three and decode any single-bit flip; use supplied 4×32 channel enumeration to compute mutual information and optimal message error and compare against the three-bit and uncoded benchmarks, accounting for rate. **Skill:** Calculate information and decision risk from one finite joint distribution without conflating information gain, code rate and guaranteed error correction. Inspected source. **Limit:** The lecture uses the lighter detection-only code; the notebook develops the stronger correcting code. Guarantee applies only to at most one flipped bit; all notebook adaptations and one-hour timing remain untested. All course adaptations remain unrun.
5	**Generalized linear models, automatic**	Start from the familiar Bayesian regression likelihood and parameter prior; explain why	• 15.1 / 15.1.1 / 15.1.3 Common model structure and derivative	• Selected 7.8.2 / 7.8.5 Matrix derivatives	Fit the three-class Iris softmax model in PyTorch; compare the manual mean-batch gradient with autograd at identical

No.	Lecture	Topics covered	Book 2 — primary	Book 1 — supplemental	Notebook problem, worked reference and skill
	differentiation and SGD Proposed teaching scope	Gaussian observations are unsuitable for binary responses and counts, without implying that the hierarchical model fails. Generalize the observation family using Gaussian/Bernoulli/Poisson GLMs and links. Contrast full posterior inference with MLE/MAP point fitting; derive the canonical NLL and vector softmax gradients, then reverse-mode autodiff and minibatch SGD.	• 15.1.2 Brief alternative-link comparison • 15.3.2 Three-class probabilities • Selected 6.2; 6.3.1–6.3.2 Automatic differentiation and batch updates	• Selected 8.2 / 8.4.1 / 8.4.3 Gradient and batch-update support	parameters, run visible SGD updates and evaluate held-out log loss/probabilities. Data preparation and program scaffolding are supplied. **Skill:** Connect a likelihood derivative with reverse-mode automatic differentiation and a checked stochastic parameter-update/evaluation loop. Inspected source. **Limit:** First PyTorch use makes this meeting tight. Data preparation and plotting are supplied; the lecture does not teach a large software framework. All course adaptations remain unrun.
6	**Latent-variable learning and expectation-maximization** Proposed teaching scope	Recall Bayesian latent-variable hierarchies and joint posterior inference, then distinguish point estimation by marginal likelihood from inference over all unknowns. Show why unknown latent values complicate direct fitting, and derive the ELBO/KL identity, exact E-step tightness and EM monotonicity. Apply Gaussian conditional first/second moments to unknown mean/covariance with missing data; briefly interpret mixture responsibilities and assumptions.	• Selected 6.5.1 / 6.5.3 Probability bound and EM • 6.5.4 Gaussian missing-data updates • 3.11 When the missing-data mechanism can be ignored • 28.2.1 Group-assignment probabilities: brief example	• Selected 3.5 / 8.7 Grouped observations and EM explanation	Assemble expected Gaussian sufficient statistics, including conditional covariance, and perform an E/M update inside a supplied iteration driver. Check observed likelihood and missing-entry reconstruction; no second full mixture fit. **Skill:** Use conditional uncertainty in sufficient-statistic estimation and distinguish an exact EM improvement guarantee from arbitrary approximate hidden-value updates. Inspected source. **Limit:** This replaces the candidate MRI tissue-classification lab. Class-assignment probabilities are explained briefly; a second mixture-model fit is not required. All course adaptations remain unrun.
7	**Bayesian computation: grid reference and Laplace approximation** Proposed teaching scope	Posterior queries and parameterization-dependent MAP; derive scalar Laplace curvature and normalizer under positive-curvature conditions. Apply to proper-prior logistic regression, check quadrature domain and resolution, and compare posterior moments and predictive integrals.	• 7.1–7.3 What posterior questions require: short introduction • Selected 7.4.1–7.4.3 Most likely parameter, grid and Gaussian shortcut • Selected 7.5 Checking approximation quality • Selected 15.3.5–15.3.6 One-parameter logistic example		Build a stable quadrature reference for the fixed eight-observation logistic posterior, challenge domain and mesh separately, and assess Laplace error in moments and prediction. Solver/plot helpers are supplied. **Skill:** Validate the numerical reference as well as the posterior approximation, and separate local curvature matching from global integral/prediction accuracy. Inspected source. **Limit:** The small example is chosen for checkability. A finite grid is also an approximation, not an automatically exact answer. All course adaptations remain unrun.
8	**Exact inference in structured sequence models** Proposed teaching scope	Derive scaled HMM forward/backward recursions, evidence and filtered/smoothed marginals; establish dense computational cost and distinguish joint path MAP from marginal modes. Give a compact Kalman-filter application of L01 Gaussian conditioning, without a second full sequence implementation.	• 9.2.1–9.2.5 Hidden-state calculations and scaling • 9.2.6 Whole history versus separate state decisions • Selected 8.1.2 / 8.2.2 Gaussian filtering: short application		Implement scaled HMM likelihood messages and marginals in NumPy. Check tiny-sequence enumeration and an asymmetric-transition case to expose orientation errors. Simulation, reference and plotting code are supplied. **Skill:** Use sum-product dynamic programming to obtain exact sequence probabilities and marginals while controlling numerical scaling. Inspected source. **Limit:** The main lab is discrete hidden-state inference. The previously preferred continuous Kalman forecasting lab is not included. All course adaptations remain unrun.
9	**Variational inference: posterior geometry and pathwise gradients** Proposed teaching scope	Reuse L06 ELBO identity; optimize restricted reverse KL and work the correlated-Gaussian diagonal optimum. Derive the total pathwise gradient including direct parameter terms and regularity conditions; apply bounded coordinate-ascent VI and recognize amortization.	• 10.1.1–10.1.2 Fitting a restricted distribution • 6.3.3 / 6.3.5; 10.2.1.1 Derivatives through Gaussian draws • 6.3.4 Another gradient estimator: statement only • 10.3.1 Coordinate updates; 10.1.5 learned inference: introduction only		Fit a diagonal Gaussian to controlled known-noise regression using reparameterized PyTorch gradients. Compare exact posterior and analytic best-diagonal solution; use fixed-noise, matched-estimator gradient checks and query-specific predictive variances. **Skill:** Separate optimization error from variational-family error, and differentiate a parameter-dependent expectation with correct full-data likelihood/prior scaling. Inspected source. **Limit:** Uses controlled data and known noise. The real

No.	Lecture	Topics covered	Book 2 — primary	Book 1 — supplemental	Notebook problem, worked reference and skill
					economic-data example and sampling comparison are replaced; no new Pyro software is required. All course adaptations remain unrun.
10	**Monte Carlo and MCMC: invariance, mixing and estimator error** Proposed teaching scope	Derive IID estimator variance, stationarity/ detailed balance and the general Metropolis–Hastings ratio, retaining rejected states. Explain Gibbs cancellation briefly and dependent-sample error under its conditions. Distinguish invariant transitions, finite-run mixing and function-specific effective sample size; recognize HMC mechanics only.	• 11.2–11.2.2 Averages and sampling error • 2.6.4; 12.2.1–12.2.2 Invariant distribution and acceptance rule • 12.3.1–12.3.2 Conditional-coordinate sampling: brief derivation • Selected 12.6.3 Dependent-sample error; 12.5.3 Hamiltonian mechanics only		Implement the NumPy MH kernel on a fixed bimodal target. Compare proposal scales and dispersed starts; assess the mean and probability above zero against exact references. Supplied diagnostics flag stuck chains and query-dependent error. **Skill:** Check transition validity separately from finite exploration and quantify uncertainty for a specified Monte Carlo query. Inspected source. **Limit:** Helper diagnostics are supplied. Gibbs updates are explained briefly. Hamiltonian Monte Carlo uses derivatives for directed moves but is not fully derived or implemented. All course adaptations remain unrun.
11	**Neural conditional models: heteroskedastic likelihood and regularization** Proposed teaching scope	Nonlinear feedforward mean/positive-scale model; derive heteroskedastic Gaussian NLL and output derivatives, reusing AD/SGD. Distinguish conditional observation noise from posterior weight uncertainty via total variance. Work the Gaussian-prior/L2 coefficient under mean-loss normalization.	• Selected 16.2.1–16.2.2 / 16.3.1 Small feedforward networks • 15.2.2.1 Gaussian prior and weight shrinkage • 14.2.1 Evaluating predicted probabilities	• 13.2.4.4 Noise that depends on input • 13.5.2 Weight shrinkage; 13.5.1 stopping training: brief mention	Fit one shallow PyTorch mean/scale network and compare held-out NLL, MSE and conditional intervals with a supplied fixed-scale nonlinear baseline and known data law. Match splits/shapes and estimate baseline noise using training data. **Skill:** Implement and interpret a neural conditional likelihood, preserving standard-deviation/ variance distinctions and the limits of fixed-weight predictive intervals. Inspected source. **Limit:** Intervals describe observation noise conditional on fitted weights. They do not include a Bayesian posterior over weights. One CPU fit and prepared fallback still need testing. All course adaptations remain unrun.
12	**Variational autoencoders: amortized inference and rate–distortion** Proposed teaching scope	Generative model versus amortized posterior; joint encoder/decoder objective with event-summed Gaussian NLL and diagonal KL, then observation averaging. Reuse ELBO/pathwise/neural results. Derive rate as mutual information plus aggregate-posterior mismatch; interpret beta, collapse and nonidentifiability.	• Selected 21.1–21.2 Model, inference network and training objective • 21.3.1 Changing the penalty weight • Selected 5.4.2 Reconstruction/ information tradeoff • Selected 21.4 When the model ignores hidden coordinates		Complete the compact mobility VAE loss and one bounded joint fit. Use fresh, consistently normalized supplied beta comparisons; evaluate expected-q reconstruction NLL and average prior KL, keeping encoder-mean visualizations separate. **Skill:** Integrate amortized variational inference with a neural likelihood and assess a defined distortion/ rate tradeoff without treating rate as mutual information or latent patterns as identified causes. Instructor mobility source. **Limit:** Uses 19 measurements and four hidden coordinates. Data access, prepared comparisons and CPU runtime remain untested. Patterns are descriptive, not identified causal factors. All course adaptations remain unrun.
13	**Project presentations and course synthesis**	Student project presentations connecting probabilistic assumptions, computation, prediction, evaluation, uncertainty and scientific interpretation. No additional substantive lecture topic is committed at this stage.			No notebook — project presentations and synthesis.

Deliberate scope boundaries

This course develops selected mathematical capabilities, not an exhaustive survey of Murphy. The exclusions below are deliberate; they are not hidden promises to cover material during L13 project presentations. The independent architecture audit evaluates source coverage and teaching feasibility.

Other possible topic	Choice in this plan	What we give up
More advanced optimization	Basic derivatives and small-batch updates are sufficient here	No separate study of curved-geometry updates, expensive black-box search, transport or subset selection.
Particle methods for changing hidden states	Exact sequence calculations in L08 and sampling theory in L10	No method for tracking a complex changing state with a cloud of weighted samples (Book 2 Ch. 13).
MRI classification and continuous forecasting	Missing-data fitting and discrete hidden-state lab selected	Less biomedical and forecasting practice. Either application can replace its corresponding lab after review.
More complex regression and prediction checks	Common models in L05 and one neural probability model in L11	No full sparse/hierarchical regression, calibration or guaranteed prediction-interval course.
Larger inference toolkits	One checked method per main lab	No general-purpose graph solver, broad variational-family catalogue or full Hamiltonian sampler derivation.
Bayesian neural networks and Gaussian processes	Small fitted networks only	No separate treatment of uncertainty about network weights or Gaussian-process prediction.
Other generative models	A variational autoencoder in L12	No flow, diffusion or adversarial model teaching. These are different ways to learn data distributions and need their own time.